STUDENT NUMBER:

APSC 171 - Term Test 2
Tuesday Nov 4, 2025

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Last Name (BLOCK LETTERS): _____

First Name: _____

Student #: _____

- This is a 2 hour test.
- No textbooks, notes, or aids other than your calculator (Casio 991 series) are allowed.
- For full marks, you must show all your work and explain how you arrived at your answers, unless explicitly told to do otherwise.
  Correct final answers with incorrect support may be given no credit.
- Wherever appropriate, **include units in your answers**.
- Keep and show at least three significant figures for decimal values.
- When drawing graphs, **add labels and scales on all axes**.
- If you need more room, there are blank pages at the end of the test. **If you use these pages, you must provide clear directions on the original page**, e.g. "Continued on page 10".
- Only answers written in this booklet will be graded; all scrap paper from the test will be discarded.

Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Total
8	6	9	6	5	5	7	9	55

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1. Evaluate the following definite integrals using the Fundamental Theorem of Calculus (i.e. using anti-derivatives).

(a) $\int_1^4 \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) dx$ [/3]

$= \int_1^4 x^{1/2} + x^{-1/2}\, dx = \frac{2}{3}x^{3/2} + 2x^{1/2} \Big|_1^4$

$= \left(\frac{2}{3}\, 4^{3/2} + 2\sqrt{4}\right) - \left(\frac{2}{3} + 2\right) = \frac{20}{3} \approx 6.67$

(b) $\int_{-2}^{-1} \left(\frac{1-x^5}{x}\right) dx = \int_{-2}^{-1} \frac{1}{x} - x^4\, dx$ [/3]

$= \ln|x| - \frac{x^5}{5} \Big|_{-2}^{-1} = \left(\ln(1) - \frac{(-1)}{5}\right) - \left(\ln(2) - \frac{(-2)^5}{5}\right)$

$= \frac{1}{5} - \ln(2) - \frac{32}{5} = -\ln(2) - \frac{31}{5}$

≈ -6.89

(c) $\int_{0.5}^{\sqrt{3}/2} \frac{1}{\sqrt{1-t^2}}\, dt = \arcsin(t) \Big|_{0.5}^{\sqrt{3}/2}$ [/2]

$= \arcsin\left(\frac{\sqrt{3}}{2}\right) - \arcsin(0.5)$

$= \pi/3 - \pi/6 = \pi/6$

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2. (a) Evaluate the integral $\int e^{(u+2)}\,du$. [/2]

$= e^{u+2} + C$

(b) Determine by differentiating whether or not your answer to part (a) is correct. State your conclusion clearly for the grader. [/1]

$\frac{d}{du}\, e^{(u+2)} + C = e^{(u+2)}$ ✓

(c) Evaluate now a second integral, $\int 3x^{4/5} - \sin(x)\,dx$. [/2]

$= 3x^{9/5} \cdot \frac{5}{9} + \cos(x) + C$

tidying $= \frac{5}{3} x^{9/5} + \cos(x) + C$

(d) Determine by differentiating whether or not your answer to part (c) is correct. State your conclusion clearly for the grader. [/1]

$\frac{d}{dx}\left(\frac{5}{3} x^{9/5} + \cos(x) + C \right)$

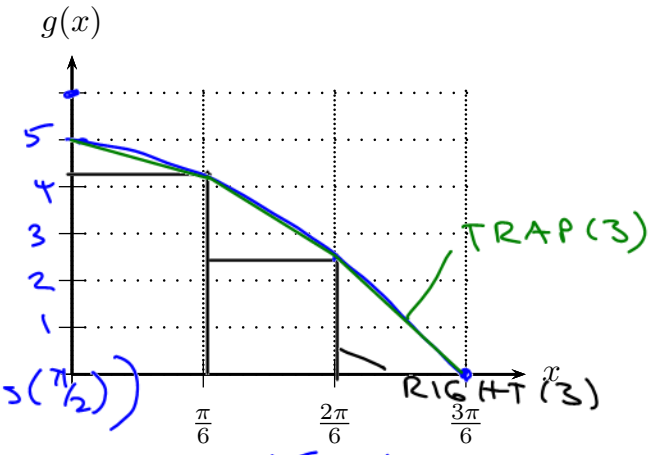
$= \frac{5}{3}\left(\frac{9}{5}\right) x^{4/5} - \sin(x)$ Matches original integrand

$= 3x^{4/5} - \sin(x)$ ✓ $\Rightarrow$ integral is correct.

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3. You are studying the function $g(x) = 5\cos(x)$ over the interval $\left[0, \frac{\pi}{2}\right]$.

(a) Sketch the graph on the axes at the right.
Clearly indicate the scale you used on the y axis. [/1]



(b) Estimate the integral $\int_0^{\pi/2} 5\cos(x)\,dx$ using a RIGHT sum with $n = 3$ intervals. Report at least 3 digits after the decimal in your answer. [/3]

RIGHT(3) = Σ (widths)(heights)

$= \frac{\pi}{6}\left(5\cos(\pi/6) + 5\cos(\pi/3) + 5\cos(\pi/2)\right)$

$= \frac{\pi}{6}\left[5\frac{\sqrt{3}}{2} + \frac{5}{2} + 0\right] \cong \boxed{3.575}$

x	$5\cos(x)$
0	5
$\pi/6$	$5\sqrt{3}/2$
$\pi/3$	$5/2$
$\pi/2$	0

(c) Estimate the integral $\int_0^{\pi/2} 5\cos(x)\,dx$ using a TRAP sum with $n = 3$ intervals. Report at least 3 digits after the decimal in your answer. [/3]

TRAP(3) = Σ (widths)(avg heights)

$= 5\cdot\frac{\pi}{6}\left(\left(\frac{1+\sqrt{3}/2}{2}\right) + \left(\frac{\sqrt{3}/2 + 1/2}{2}\right) + \left(\frac{1/2 + 0}{2}\right)\right) = 5\cdot\frac{\pi}{6}\left(\frac{2+\sqrt{3}}{2}\right) \cong \boxed{4.885}$

(d) Use the Fundamental Theorem of Calculus to compute the exact value of the integral $\int_0^{\pi/2} 5\cos(x)\,dx$. [/2]

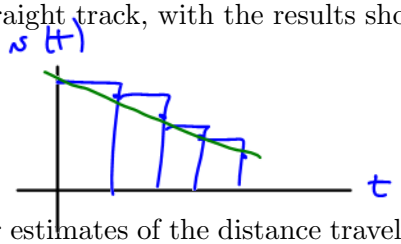
$\int_0^{\pi/2} 5\cos(x)\,dx = 5\sin(x)\Big|_0^{\pi/2} = 5\left(\sin(\pi/2) - \sin 0\right)$

$= 5(1-0) = \boxed{5}$

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4. A radar tracks the velocity of a car as it is driving along a straight track, with the results shown in the table below.

Time (s)	0	0.25	0.5	0.75	1.0
Velocity (m/s)	12.3	10.8	9.5	8.4	7.5



(a) Use LEFT and RIGHT sums to compute upper and lower estimates of the distance travelled by the car over the period $t = 0 \ldots 1$. [/3]

$LEFT(4) = 0.25\,[12.3 + 10.8 + 9.5 + 8.4]$

$= 10.25$ m

$RIGHT(4) = 0.25\,[10.8 + 9.5 + 8.4 + 7.5]$

$= 9.05$ m

Upper distance estimate: 10.25 m

Lower distance estimate: 9.05 m

(b) Using function fitting, you find that the function $v(t) = 1.6t^2 - 6.4t + 12.3$ closely approximates the velocity values recorded by the radar. Using this estimated function, compute another estimate of the total distance travelled by the car over the period $t = 0 \ldots 1$. [/2]

$$\text{distance} = \int_0^1 v(t)\,dt = \int_0^1 1.6t^2 - 6.4t + 12.3\,dt$$

$$= \frac{1.6t^3}{3} - \frac{6.4}{2}t^2 + 12.3t \Big|_0^1$$

$$= \frac{1.6}{3} - 3.2 + 12.3 = 9.6\dot{3}$$

distance estimate: 9.63 m

(c) Assuming that the car's motion is reasonably smooth in between the recorded data points, estimate the **acceleration** of the car at $t = 0.75$ s. Several estimates are possible, and any reasonable method will be accepted. [/1]

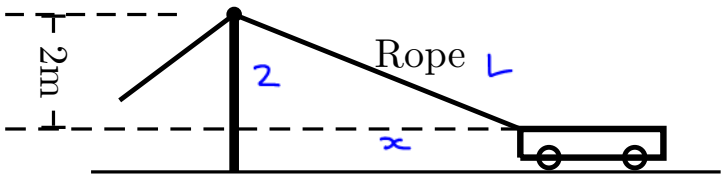
Any of these:

- derivative of $v(t)$: $a(t) \cong 3.2t - 6.4$; at $t = 0.75$, $a \cong -4$ m/s^2
- $a(t) \approx \frac{\Delta v}{\Delta t}$ from table: $\frac{7.5 - 8.4}{0.25} = -3.6$ m/s^2 or $\frac{7.5 - 9.5}{2(0.25)} = -4$ m/s^2

or $\frac{8.4 - 9.5}{0.25} = -4.4$ m/s^2

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5. A cart is being pulled in towards a pole by a rope attached to the front of the cart and passing through a pulley. The pulley is on the pole and 2 m higher than the rope mounting point on the cart.
If the rope is pulled in at a rate of 0.8 m/s, how fast is the cart approaching the pole when the closest part of the cart is 9 m from the pole?



[/5]

Know: $\frac{dL}{dt} = -0.8$ m/s ; want $\frac{dx}{dt}$

Always true: $x^2 + 2^2 = L^2$ ①

Take $\frac{d}{dt}$ of both sides: $\frac{d}{dt}(x^2 + 4) = \frac{d}{dt}(L^2)$

$$2x\frac{dx}{dt} = 2L\frac{dL}{dt}$$

Solve for $\frac{dx}{dt}$: $\frac{dx}{dt} = \frac{2L}{2x}\frac{dL}{dt}$

When $x = 9$, also need L: ① → $(9)^2 + 2^2 = L^2$

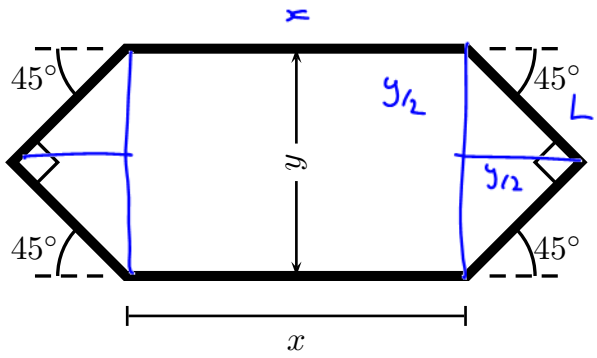
$L = \sqrt{85} \approx 9.22$

$\Rightarrow \frac{dx}{dt} = \frac{9.22}{9}(-0.8) \approx -0.82$ m/s

Cart is approaching the pole at 0.82 m/s

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6. Consider the symmetric diamond-shaped region shown to the right.
The perimeter of the region will be made with 100 m of fencing material. Find the lengths x and y, in meters, that result in the largest possible enclosed area, and compute that area. Record your final values where indicated at the bottom of the page, to 2 digits after the decimal.
You do **not** need to show your answer is a local maximum for the area.



[/5]

$L^2 = (y/2)^2 + (y/2)^2$

$L^2 = 2\left(\frac{y^2}{4}\right) = \frac{y^2}{2}$

$L = \sqrt{\frac{y^2}{2}} = \frac{y}{\sqrt{2}}$

Perimeter $= 2x + 4L$

$= 2x + 4\left(\frac{y}{\sqrt{2}}\right) \overset{\text{Set}}{=} 100$

Isolating $x \Rightarrow \quad 2x = 100 - \frac{4y}{\sqrt{2}}$

$\boxed{x = 50 - \frac{2y}{\sqrt{2}}}$ (1)

Area $=$ $\frac{y}{2}$ (triangle) $+$ $y \times x$ (rectangle) $+$ $\frac{y}{2}$ (triangle)

$= \frac{1}{2}(y)\left(\frac{y}{2}\right) + xy + \frac{1}{2}(y)\left(\frac{y}{2}\right)$

$= 2\frac{y^2}{4} + xy = \frac{y^2}{2} + xy$ (1)

$A(y) = \frac{y^2}{2} + \left(50 - \frac{2y}{\sqrt{2}}\right)y$

Find critical points:

$\frac{d}{dy}A = \frac{2y}{2} + 50 - \frac{4y}{\sqrt{2}} \overset{\text{Set}}{=} 0$

$50 = -y + \frac{4y}{\sqrt{2}} = y\left(-1 + \frac{4}{\sqrt{2}}\right)$

$y = \frac{50}{\left(4/\sqrt{2} - 1\right)} \approx 27.35$ m

$\Rightarrow \quad x = 50 - \frac{2y}{\sqrt{2}} \approx 11.32$ m

$\Rightarrow A = \frac{y^2}{2} + xy \approx 683.46\ m^2$

Optimal x: 11.32 m

Optimal y: 27.35 m

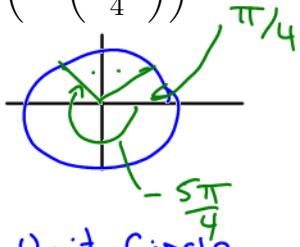
Largest area: 683.46 m^2

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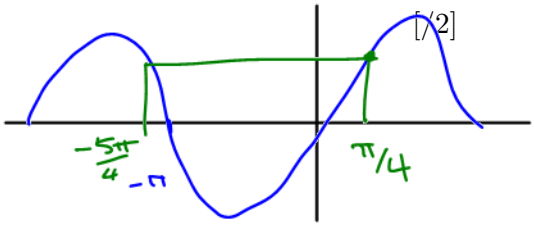
7. The following three parts are independent of each other.

(a) Find the exact value of $x = \arcsin\left(\sin\left(\frac{-5\pi}{4}\right)\right)$. [/2]

$x = \pi/4$



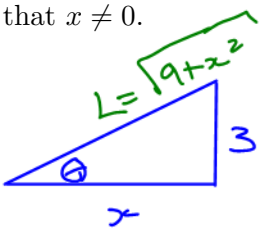
Unit Circle



Sine graph

(b) Express the relationship $y = \sin\left(\arctan\left(\frac{3}{x}\right)\right)$ without trigonometric functions. You may assume that $x \neq 0$. [/2]

let $= \theta$



$L^2 = 3^2 + x^2 \rightarrow L = \sqrt{9+x^2}$

$\sin(\theta) = \dfrac{3}{\sqrt{9+x^2}}$

(c) The CPU in a computer has a recommended clock speed or frequency, but can be "over-clocked" to run faster (process more instructions per second). One of the downsides of over-clocking is that the CPU uses more electrical power.

We will define $P(f)$ as the function that maps the clock speed or frequency f, in GHz, to the power used, in Watts, to run a particular CPU at full load.

An Intel Core i7-5960X has a recommended clock speed of $f = 3.5$ GHz. Through experiments you find that $P(3.5) = 205$, with the frequency in GHz and the power in Watts. You also find that $P'(3.5) = 95$.

(i) What are the units of $P'(3.5)$? [/1]

P' units are W/GHz.

(ii) Estimate the power required to run the CPU at $f = 3.8$ GHz. [/2]

$P(f) \approx P(3.5) + P'(3.5)(f - 3.5)$

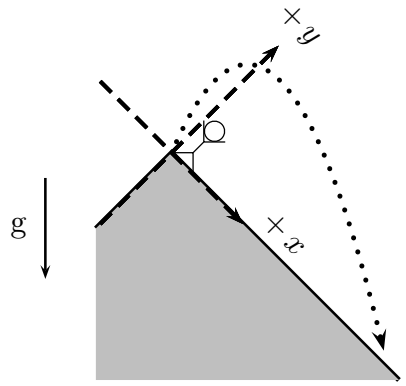
$P(3.8) \approx 205 + 95(3.8 - 3.5)$

$= 233.5$ Watts

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8. You throw a ball down the side of a hill. Relative to the xy coordinate axes shown in the diagram, the velocity of the ball at time t (in seconds) is given by

$$\vec{v}(t) = \langle \underbrace{7t-2}_{v_x(t)}, \underbrace{-7t+5}_{v_y(t)} \rangle \text{ m/s.}$$



(a) Give a function $\vec{r}(t) = \langle x(t), y(t) \rangle$ which describes the position of the ball at time t. Assume that the ball is thrown from $\langle x, y \rangle = \langle 0, 0 \rangle$ at $t = 0$. [/3]

$\vec{r}(t) = \left\langle \frac{7t^2}{2} - 2t + c_1, -\frac{7t^2}{2} + 5t + c_2 \right\rangle$

Since $\vec{r}(0) = \langle 0,0 \rangle$, both $c_1 = c_2 = 0$.

$\Rightarrow \vec{r}(t) = \left\langle \frac{7t^2}{2} - 2t, -\frac{7t^2}{2} + 5t \right\rangle$

(b) At what time t will the ball hit the hillside? [/2]

$\Rightarrow$ y coordinate $= 0 \Rightarrow 0 = -\frac{7t^2}{2} + 5t$

$0 = t\left(-\frac{7}{2}t + 5\right)$

$t = 0$ or $-\frac{7}{2}t + 5 = 0 \rightarrow t = 5 \cdot \frac{2}{7} = \frac{10}{7} \approx 1.429$ s

launch x

(c) Since there is no wind, the only acceleration on the ball is due to gravity. Give a vector in xy coordinates which points in the direction of gravity. [/2]

$\vec{a}(t) = \frac{d}{dt}\vec{v}(t) = \langle 7, -7 \rangle$

(d) The gravity on Earth, Mars and Neptune are 9.8m/s^2, 3.8m/s^2 and 13.3m/s^2 respectively. Which planet are we on? Be sure to justify your answer. [/2]

$\|\vec{a}(t)\| = \|\langle 7, -7 \rangle\| = \sqrt{7^2 + 7^2} = \sqrt{98} \approx 9.89 \text{ m/s}^2$

We are on Earth.

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Space for additional work. **Indicate clearly which question you are continuing if you use this space.**

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